

Arthropod Born Viral Zoonoses

Sr. No.	Family of the virus	Genus	In animals		In Humans		Confirmatory test
			Disease	Samples to be collected	Disease	Samples to be collected	
1	Flaviviridae	Flavivirus (JE virus)	Encephalitis	Blood , brain, CSF, serum	Japanese encephalitis	Blood , brain, CSF, serum	Virus isolation (mosquito cell line), IgM capture ELISA, CFT, HI,
2		Flavivirus (St. Louis virus)	-	Blood	St. Louis encephalitis	Blood	ELISA, CFT, HI
3		Flavivirus (West Nile virus)	-	Blood, CSF, brain, serum	West Nile encephalitis	Blood, CSF, brain, serum	Virus isolation, suckling mice inoculation, ELISA, CFT, HI
4		Flavivirus (Yellow Fever virus)	Yellow fever	Blood, serum	Haemorrhagic fever, black vomit	Blood , serum	Isolation in mice, Rh monkeys, tissue culture, CFT, ELISA, HI,
5		Flavivirus (KFD virus) (TBE virus)	Kayasanur forest disease, Monkey disease	Blood, serum	Tick borne encephalitis Kayasanur forest disease	Blood, serum	Mouse inoculation, CFT, ELISA, HI
6		Flavivirus (Louping ill virus)	Louping ill	Blood, CSF, brain, serum	Encephalitis	Blood, CSF, brain, serum	Virus isolation on sheep kidney cell lines, chicken embryo/mouse inoculation, CFT, ELISA, HI, serum neutralization
7		Flavivirus (OMSK HF)	Haemorrhagic disease in	Blood, serum	OMSK Haemorrhagic	Blood, serum	Virus neutralization,

			rodents		fever		ELISA, CFT, HI
8		Flavivirus (Powassan virus)	Subclinical infections	Serum	Powassan encephalitis	Serum, brain	Isolation by intracerebral inoculation in mice, serum neutralization, HI
9		Flavivirus (Sub type RSSE)	Meningo encephalitis	Blood, serum	Russian Spring Summer Encephalitis	Blood, serum	ELISA, CFT, Serum neutralization
11	Bunyaviridae	Bunya virus	Encephalitis	Blood, serum	California encephalitis	Blood, serum, brain	ELISA, CFT, virus neutralization, IgM capture EISA, isolation from brain
12		Phlebovirus (RVF virus)	-	Blood, liver, serum	Rift Valley Fever, Enzootic hepatitis	Blood, liver, serum	Isolation in mice, ELISA, CFT, Histopathology of liver
13		Nairovirus (Ganjam virus)	-	Blood, serum	Haemorrhagic fever	Blood, serum	CFT, ELISA, Serum neutralization
14		Bhanja virus	-	Blood, serum	Haemorrhagic fever	Blood, serum	CFT, ELISA, virus neutralization
15	Togaviridae	Alpha virus (EEE virus)	Eastern equine encephalitis	Brain, blood	Encephalitis	Brain, blood	Isolation from brain, CFT, HI, serum neutralization
16		Alpha virus (WEE virus)	Western equine encephalitis	Brain ,serum	Encephalitis	Brain, serum	Isolation from brain, HI, CFT
17		Alpha virus	Venezuelan	Nasopharyngeal	Venezuelan	Nasopharyngeal	CFT, HI

		(VEE)	Equine Encephalitis	exudate, blood	Encephalitis	exudate, blood, serum	
18		Alpha virus (Chicken Gunya virus)	-	Blood, serum	Chicken Gunya	Febrile blood, serum	CFT, HI, serum neutralization, isolation from febrile blood in suckling mice, vero cell lines
19		Alpha virus (Mayaro virus)	-	Blood, serum	Mayaro Fever	Blood, serum	Isolation in new born mice, CFT, HI
20		Alpha virus (Sindhbis virus)	-	Blood, serum	Sindhbis Fever	Blood, paired serum	Isolation by mice inoculation, HI, CFT
21		Alpha virus (Ross River virus)	-	Blood, serum	Epidemic poly arthritis, Ross River Fever	Blood, paired serum	CFT (four fold rise in antibody titre), HI, isolation by mice inoculation, IFT
22	Reoviridae	Colti virus (Colorado Tick Fever Virus)	-	Blood, serum	Colorado Tick Fever	Blood serum	CFT, Immuno fluorescence

Non Arthropod born viral Zoonoses

Sr. No.	Family of the virus	Genus	In animals		In Humans		Confirmatory test
			Disease	Samples to be collected	Disease	Samples to be collected	
1	Orthomyxoviridae	Influenza A virus	Influenza/Grippe	Nasal swab, throat swab, paired serum	Influenza	Nasal swab, throat swab, paired serum, sputum	Isolation in hamsters/mice, CFT (four fold rise in the antibody titre), HI, ELISA, AGPT
2	Paramyxoviridae	Paramyxo virus	Nipah virus infection	Paired serum, blood	Nipah virus infection	Paired serum, blood	ELISA, HI, serum neutralization, isolation in chicken embryo/ tissue culture
3		Paramyxo virus	Hendra virus infection	Paired serum, blood	Hendra virus infection	Paired serum, blood	ELISA, HI, serum neutralization, isolation in chicken embryo
4		Paramyxo virus (New Castle Disease virus)	New Castle Disease	Blood, serum	Conjunctivitis	Blood, serum	ELISA, HI, serum neutralization, isolation in chicken embryo/ tissue culture
5	Filoviridae	Filovirus (Marburg and Marburg like viruses)	Marburg virus disease , Green monkey disease	Blood, paired serum	Haemorrhagic fever encephalitis	Vomit, feces, blood, paired serum	Isolation on tissue culture,
6		Filovirus (Ebola & Ebola like	African	Blood,	Haemorrhagic	Blood,	Indirect immuno

		viruses)	Haemorrhagic fever Ebola virus haemorrhagic fever	serum,	fever encephalitis	serum	electron microscopy, ELISA, RTPCR
7	Arenaviridae	Arena virus	Argentine haemorrhagic fever	Blood, serum	Haemorrhagic fever encephalitis	Febrile blood, serum	IgG and IgM ELISA, Indirect immunofluorescence, RTPCR, isolation by injecting in suckling mice
8		Arena virus (Lassa virus)	Lassa fever	Blood, pharyngeal washing swab, serum	Haemorrhagic fever	Blood, pharyngeal washing swab, serum	CFT, serum neutralization, HI, virus isolation on tissue culture
9		Arena virus (LCM)	Lymphocytic choriomeningitis	CSF, brain, blood, lung, viscera, serum	Meningitis,	CSF, brain, blood, lung, viscera, paired serum	Isolation in guinea pig/mice/tissue culture, CFT (four fold rise in antibody titer), RTPCR, Immunofluorescence
10		Arena virus (Machupo)	Bolivian haemorrhagic fever		Haemorrhagic fever encephalitis		
11		Arena virus (Sabia)	Inapparent infection in animals	Difficult to diagnose	Haemorrhagic fever encephalitis	Febrile blood	Isolation of virus
13	Flaviviridae	Flavivirus (Bat salivary gland fever virus)	Bat Salivary Galnd Fever	Blood, serum	Haemorrhagic fever encephalitis	CSF, blood, serum	Isolation of virus, electron microscopy, CFT, virus neutralization,

							ELISA
14	Rhabdoviridae	Lyssavirus (genotype 1-7)	Rabies	Saliva, CSF, brain, salivary gland, corneal smear, blood, serum	Rabies	Saliva, CSF, brain, salivary gland, corneal smear, blood, serum	Isolation of virus in weaning mice, electron microscopy, CFT, virus neutralization, CIEP, ELISA, Histopathological examination (negri bodies)
		Vesiculo virus	Vesicular stomatitis	Blood, serum, vesicular fluid	Vesicular stomatitis	Blood, serum	CFT, HI in humans, virus isolation in animals
16		Vesiculo virus (Chandipura virus)	Chandipura Encephalitis	CSF, blood, serum	Chandipura Encephalitis	CSF, blood, serum	CFT, ELISA, virus isolation in animals
17	Herpesviridae	Herpes simiae	Herpes infection	Blood, serum	Herpes infection	Blood, brain, serum	Isolation from medulla oblongata, serum neutralization test
18		Herpes simplex type 1	Vesicular eruptions	Blood, vesicular fluid, serum	Vesicular eruptions	Blood, vesicular fluid, serum	Isolation of virus from clinical specimen, ELISA, serum neutralization
19	Poxviridae	Buffalo pox virus	Pox	Blood, pox lesions, serum	Pox	Blood, pox lesions, serum	ELISA, AGID, isolation on chicken embryo (pock formation)
20		Camel pox virus	Pox	Blood, pox lesions, serum	Pox	Blood, pox lesions, serum	ELISA, AGID, isolation on chicken embryo (pock

							formation)
21		Cow pox virus	Pox	Blood, pox lesions, serum	Pox	Blood, pox lesions, serum	ELISA, AGID, isolation on chicken embryo (pock formation)
22		Monkey pox virus	Pox	Blood, pox lesions, serum	Pox	Blood, pox lesions, serum	ELISA, AGID, isolation on chicken embryo (pock formation), intra cytoplasmic intrusion bodies
23		Parapox viruses (orf T)	Orf	Blood, serum, vesicular lesions	Stomatitis	Blood, serum, vesicular lesions	ELISA, AGID, isolation on chicken embryo (pock formation)
24		Bovine popular virus	Bovine popular stomatitis	Blood, lesions, serum	Popular lesions in mouth and stomach	Blood, lesions	Isolation of kidney, bovine testicles, tissue culture, electron microscopy, ELISA, RTPCR
25		Pseudo cow pox virus	Skin pox lesions	Blood, serum, vesicular fluid	Pox lesions	Blood, vesicular fluid, serum	Isolation of bovine kidney cell lines, electron microscopy, ELISA
26		Yabapox viruses	Yaba monkey tumor	Blood, serum, vesicular lesions	Skin pox lesions	Blood, serum, vesicular lesions	Isolation of virus in tissue culture, ELISA
27		Tana pox virus	Skin pox lesions	Blood, serum, vesicular lesions	Skin pox lesions	Blood, serum, vesicular lesions	Isolation of virus in tissue culture, ELISA

28	Pircornaviridae	Apthus virus	FMD	Vesicular lesions, blood, paired serum	Blister, vesicles on surface of finger palm of hand, on toe/feet etc.	Vesicular lesions, blood, paired serum	Isolation by inoculation in suckling mice, tissue culture, lesions will develop on mice, CFT (rise in antibody titer), ELISA, RTPCR
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